

- 1 Which of the following is true about organelles?
- A Vesicles from the smooth endoplasmic reticulum carry lipids to the Golgi apparatus where they are modified.
 - B Mitochondria and chloroplasts can be seen under a light microscope.
 - C Mitochondria share the same 80s ribosome as the prokaryotic cell.
 - D Hydrolytic enzymes are found in lysosomes in animal and plant cells.

- 2 With reference to the table below, which of the following combinations of bonds correctly contributes to the structure of molecules shown?

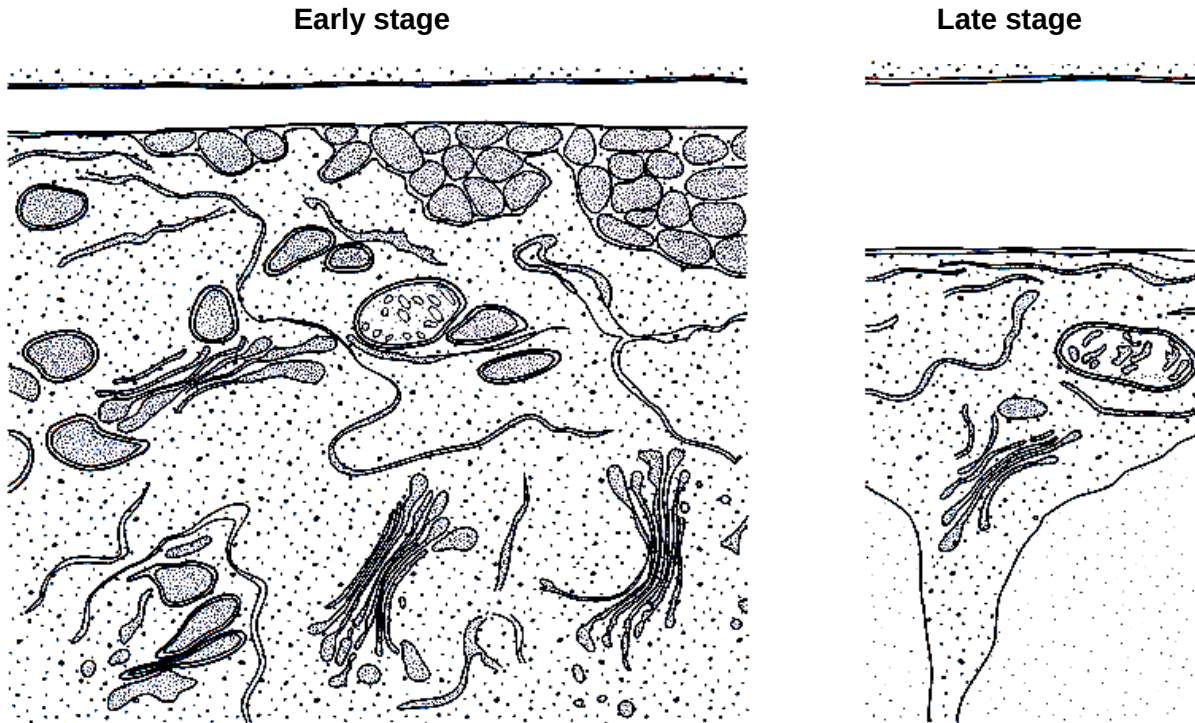
	Amylase	Cellulose	Glycogen	Insulin
A	α 1,4 glycosidic bond	β 1,4 glycosidic bond	α 1,6 glycosidic bond	hydrogen bond
B	α 1,6 glycosidic bond	hydrogen bond	α 1,4 glycosidic bond	ester bond
C	covalent bond	hydrogen bond	α 1,6 glycosidic bond	phosphodiester bond
D	ionic bond	β 1,4 glycosidic bond	α 1,4 glycosidic bond	peptide bond

- 3 The **R** group of the amino acid *serine* is $-\text{CH}_2-\text{OH}$. The **R** group of the amino acid *alanine* is $-\text{CH}_3$.

Where would you expect to find these amino acids in a globular protein in aqueous solution?

- A Serine would be in the interior, and alanine would be on the exterior of the globular protein.
- B Alanine would be in the interior, and serine would be on the exterior of the globular protein.
- C Both serine and alanine would be in the interior of the globular protein.
- D Both serine and alanine would be in the interior and on the exterior of the globular protein.

- 4 The photo electron micrographs show early and late stages in the development of the cell wall in a young plant cell.

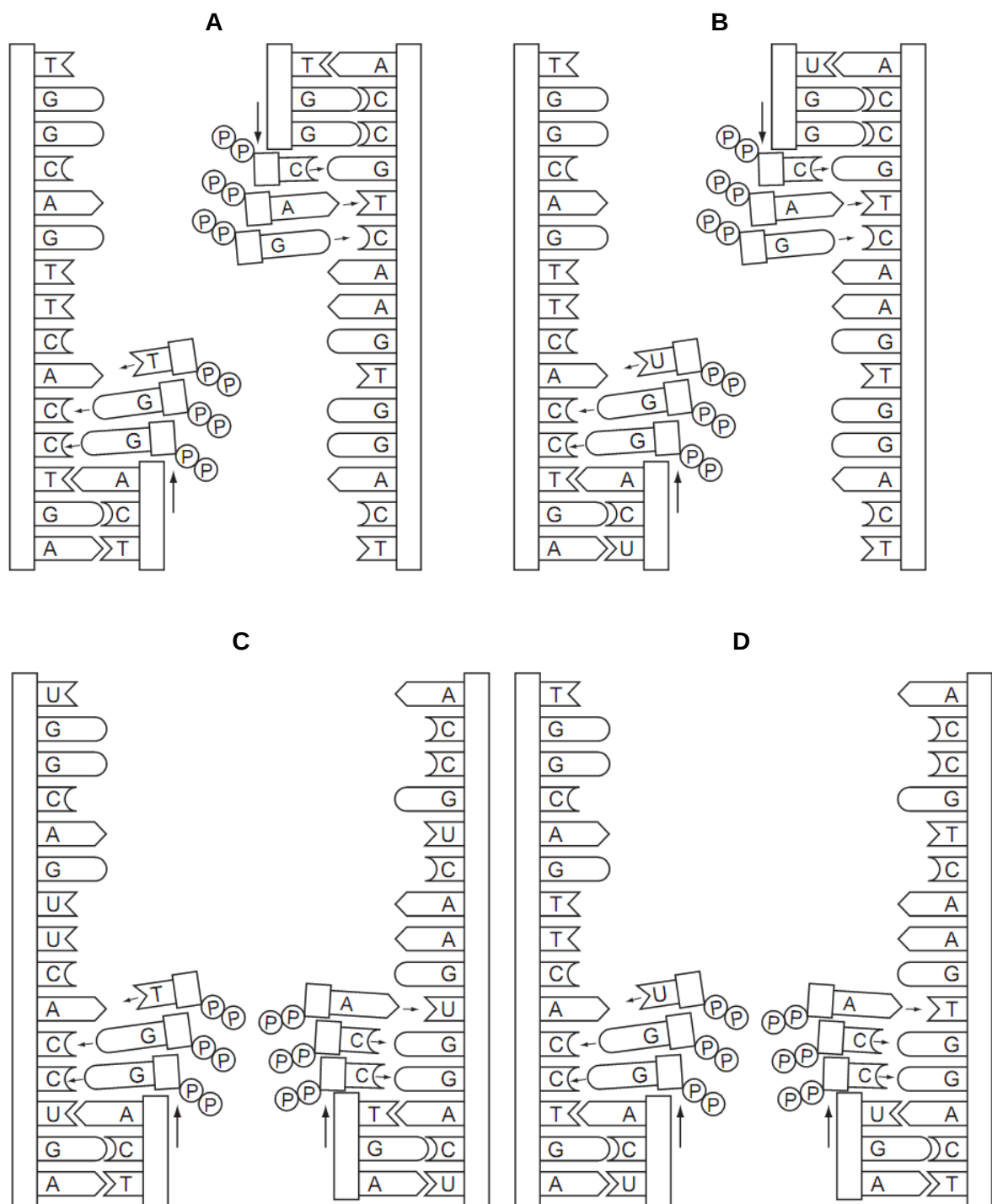


Which statement describes the events leading to the development of the cell wall?

- A Complex carbohydrates assembled in the Golgi body are exported to the cell wall by the Golgi vesicles.
- B Enzymes in the cell surface membrane synthesise the cell wall components from soluble carbohydrates brought by the Golgi vesicles.
- C Polysaccharides are exported to the cell wall and synthesized into cell wall components by the Golgi body.
- D Ribosomes synthesise glycoproteins that are exported by Golgi vesicles to be used in the cell wall.

- 5 When investigating enzyme/substrate interactions, which one of the following would be expected to show a linear relationship under constant conditions?
- A Amount of product against time, with the amount of substrate limited.
 - B Rate of reaction against substrate concentration, with the amount of enzyme limited.
 - C Rate of reaction against enzyme concentration, with the amount of substrate limited.
 - D Rate of reaction against enzyme concentration, in the presence of excess substrate.
- 6 Which of the following statement is true for enzyme inhibitors?
- 1 Inhibitors that form covalent bonds with enzymes often inactivate the enzymes permanently.
 - 2 They often target enzymes at the beginning of a metabolic pathway in end product inhibition.
 - 3 Enzyme inhibitors repress the transcription of the genes coding for enzymes.
 - 4 Post translational modifications are an effective way to regulate the activity of the inhibitors.
- A 2 and 4
 - B 2, 3 and 4
 - C 1, 2 and 4
 - D 1, 2, 3 and 4
- 7 A cell in the G1 phase has two homologous pairs of chromosomes. It undergoes a mitotic division, followed by meiosis.
- At the end of meiosis 2, what is the sum of all the products of these divisions in all cells?
- A 8 chromosomes, 16 chromatids, 16 cells, 4 times as many alleles.
 - B 8 chromosomes, 16 chromatids, 8 cells, 2 times as many alleles.
 - C 16 chromosomes, 16 chromatids, 8 cells, 4 times as many alleles.
 - D 16 chromosomes, 16 chromatids, 4 cells, 4 times as many alleles.

8 Which diagram shows the semi-conservative replication of a section of a molecule of DNA?



- 9 A transcription unit 8000 nucleotides long may use 1200 nucleotides to make a protein consisting of 400 amino acids.

Which of the following best explains the phenomenon?

- A Many non-coding nucleotides are present in mature mRNA.
 - B Many nucleotides are needed to code for each amino acid.
 - C Nucleotides break off and are lost during the transcription process.
 - D There are introns in the gene that are the noncoding sequences.
- 10 The diagram shows the first five codons of an mRNA molecule and the first five amino acids of the encoded polypeptide.

AUG - AAG - UGU - UAU - CUC - AUG
Methionine-Lysine-Cysteine-Tyrosine-Leucine-Methionine

As a result of a mutation, a dipeptide was produced instead of the normal polypeptide. The triplets on the DNA strand that is complementary to mRNA, that code for 'stop', are ATT, ATC and ACT.

Which statement explains the production of a dipeptide?

- A The DNA triplet coding for methionine is changed from TAC to ATC.
 - B The DNA triplet coding for lysine is changed from TTC to ATC.
 - C The DNA triplet coding for cysteine is changed from ACA to ACT.
 - D The DNA triplet coding for tyrosine is changed form ATA to ATG.
- 11 In birds, sex is determined by a ZW chromosome scheme. Males are ZZ and females are ZW. A recessive lethal allele that causes death of the embryo is sometimes present on the Z chromosome in pigeons. What would be the sex ratio in the offspring of a cross between a male that is heterozygous for the lethal allele and a normal female?
- A 2:1 male to female
 - B 1:2 male to female
 - C 1:1 male to female
 - D 3:1 male to female

- 12** Victoria's father has blood group AB while her mother's blood group is unknown. The blood group of Victoria's husband, Albert, is also not known. Albert's mother has blood group O. Victoria and Albert plan to have a child.

Which of the following deductions about the blood group of their first child is **not** likely?

- A** The child can have a blood group different from that of Albert.
- B** The child can have blood group A, B, AB or O.
- C** The child can have a blood group different from that of Victoria.
- D** The child cannot have blood group O.
- 13** In corn, the colour of the stem can be either wild type, which is dark green, or virescent, which is pale green.

Cross	Phenotype of parents	Temperature at which raised	Offspring
1	virescent X virescent	20°C	virescent
2	virescent X virescent	37°C	wild type
3	wild type X wild type	20°C	3 wild type: 1 virescent
4	wild type X wild type	37°C	wild type

The parents of crosses 1 and 2 had the same genotypes. The parents of crosses 3 and 4 had the same genotypes.

Which of the following can be deduced from the data?

- (i) Wild type is dominant.
- (ii) The wild type parents are heterozygotes.
- (iii) The virescent phenotype is recessive but the characteristic will only be expressed at a lower temperature.
- (iv) The parents of cross 1 are only recessive at 20°C.

- A** (ii) only
- B** (i) and (iii) only
- C** (i), (ii) and (iii) only
- D** (i), (ii), (iii) and (iv)

- 14** Aberrations during crossing over in Prophase I of meiosis may result in chromosomal mutations. Which of the following mutations is least likely to result?

A Deletion
B Polyploidy
C Duplication
D Translocation

- 15** In an experiment, chloroplast extracts were first treated with a chemical that 'snatches' away the electron that was accepted by the electron acceptor in photosystem I. The extracts were then treated with 2 hours of light and were provided with ample carbon dioxide and water.

Which of the following correctly shows the products that were formed after the experiment?

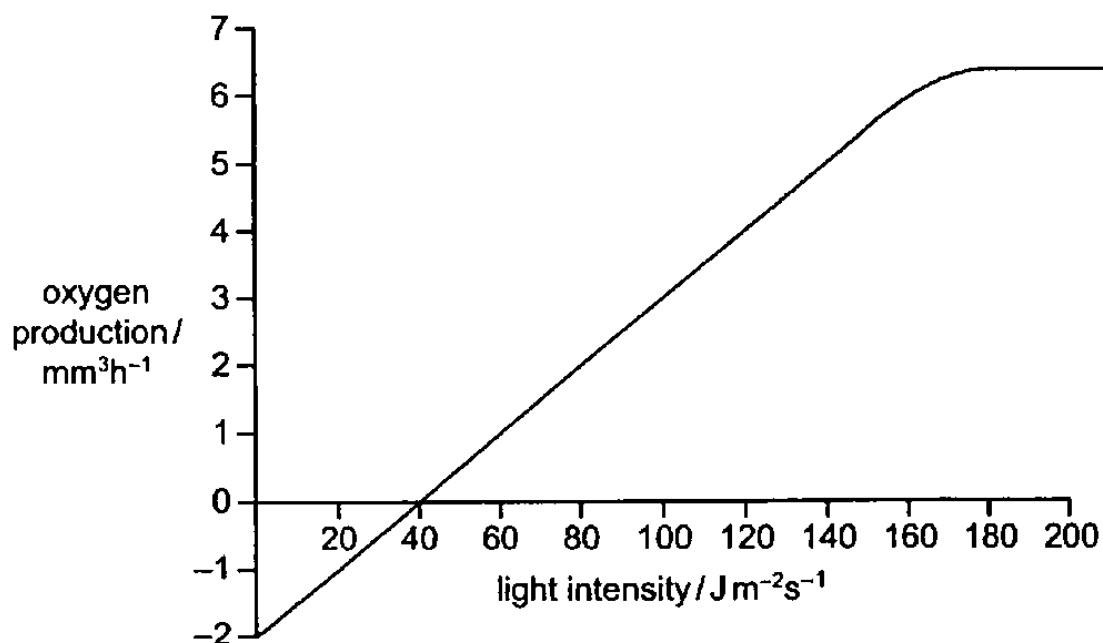
	O₂	ATP	NADPH	Triose phosphate
A	+	+	–	–
B	–	+	+	+
C	+	–	–	–
D	–	–	+	+

Key: (+) = present, (–) = absent

- 16** Which statement about anaerobic respiration is correct?

A Animals are unable to use lactate for the production of ATP.
B From one molecule of glucose, ethanol and lactate production yield the same amounts of ATP.
C From one molecule of glucose, ethanol production yields more energy than lactate production.
D Yeast is able to respire ethanol for the production of ATP.

- 17 The graph shows the relationship between oxygen production in photosynthesis and light intensity for a unicellular green organism in 0.02% sodium hydrogencarbonate solution.



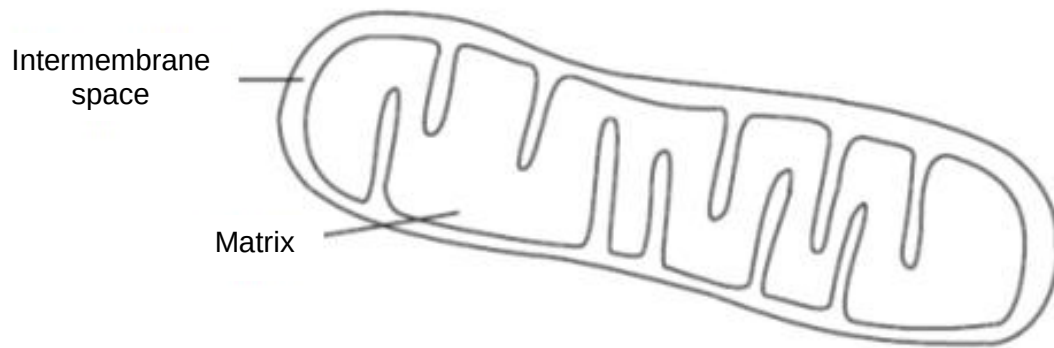
What is the most likely explanation for the graph leveling off at $180 \text{ J m}^{-2} \text{s}^{-1}$?

- A light limited and carbon dioxide saturated
 - B light limited and the temperature is below optimum
 - C light saturated and carbon dioxide limited
 - D light saturated and the temperature is above optimum
- 18 An inhibitor of a Krebs cycle enzyme that catalyses the oxidation of citrate is added to a respiring tissue.

What would be the effect of the inhibitor on the concentration of oxaloacetate, pyruvate and ATP in the respiring tissue?

	oxaloacetate	pyruvate	ATP
A	decreases	increases	decreases
B	decreases	decreases	increases
C	increases	increases	decreases
D	increases	decreases	increases

- 19** The diagram below shows a mitochondrion as seen under the electron microscope.



What is the advantage of having a small volume inside the intermembrane space of the mitochondrion?

- A** A high electron concentration is rapidly developed.
 - B** A high proton concentration is rapidly developed.
 - C** Enzymes of the Krebs cycle are highly concentrated.
 - D** Protein electron carriers are highly concentrated.
- 20** Which one of the following sentences regarding the transport of substances across the membrane is true?
- a** In diffusion, there will be no movement of particles across the membrane once dynamic equilibrium is reached.
 - B** Both diffusion and osmosis can occur in the presence of membranes and the absence of ATP.
 - C** In order for a cell to reach the same water potential as its surrounding solution, there must be no net gain or loss of water.
 - D** The sodium-potassium pump is a channel protein that is associated with the facilitated diffusion of sodium and potassium ions across the plasma membrane.

- 21** In the summer of 1995, at least 15 iguanas survived Hurricane Marilyn on a raft of uprooted trees. The rest of the mainland population was wiped out by the hurricane. These 15 iguanas rode the high seas for a month before colonizing the Caribbean island, Anguilla. These few individuals were perhaps the first of their species, *Iguana iguana*, to reach the island.

Which of the following best explains this?

- A** Natural selection
- B** Artificial selection
- C** Founder effect
- D** Bottleneck effect

- 22** Which of the following ideas were **not** part of Darwin's theory of evolution by natural selection?

- (i) Acquired variation is inherited.
- (ii) Variation between individuals arises by gene mutation.
- (iii) Organisms produce more offspring than the environment can support.
- (iv) Only those individuals best adapted to the environment survive and reproduce.
- (v) Regions that encode portions of the polypeptide that are vital for structure and function are less likely to incur mutations.

- A** (i), (ii) and (v)
 - B** (i), (iii) and (iv)
 - C** (ii), (iii) and (v)
 - D** (ii), (iv) and (v)
- 23** Molecular homology of myoglobin is used to determine evolutionary relatedness in a range of mammals. Either nucleic acids or proteins can be sequenced. The sequence of nucleic acids and proteins are compared.

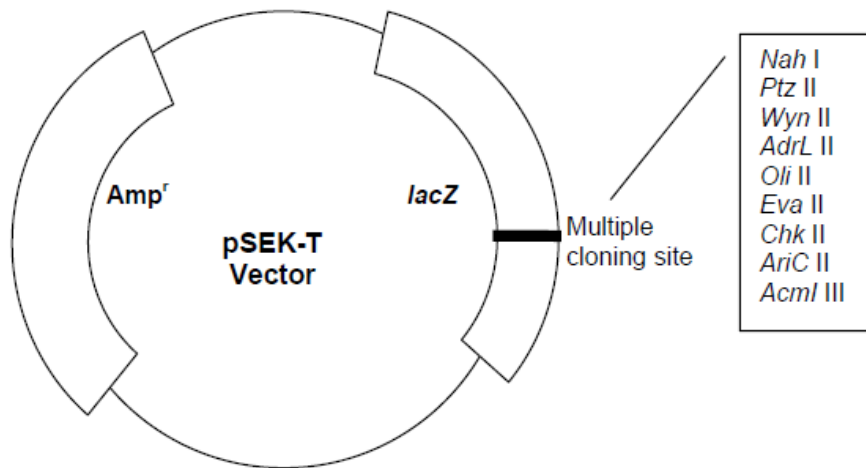
Which sequence will vary least in related mammals?

- A** DNA
- B** primary mRNA transcript
- C** edited mRNA transcript
- D** amino acid

- 24 The wounds of patients who have undergone surgery are susceptible to bacterial infections by *Enterococcus faecium* and methicillin-resistant *Staphylococcus aureus* (MRSA). Methicillin and other antibiotics are often used to treat infections of surgical wounds, but do not always cure them.

Which of the following statements about the development of resistance to antibiotics in bacteria such as *Enterococcus faecium* and MRSA is correct?

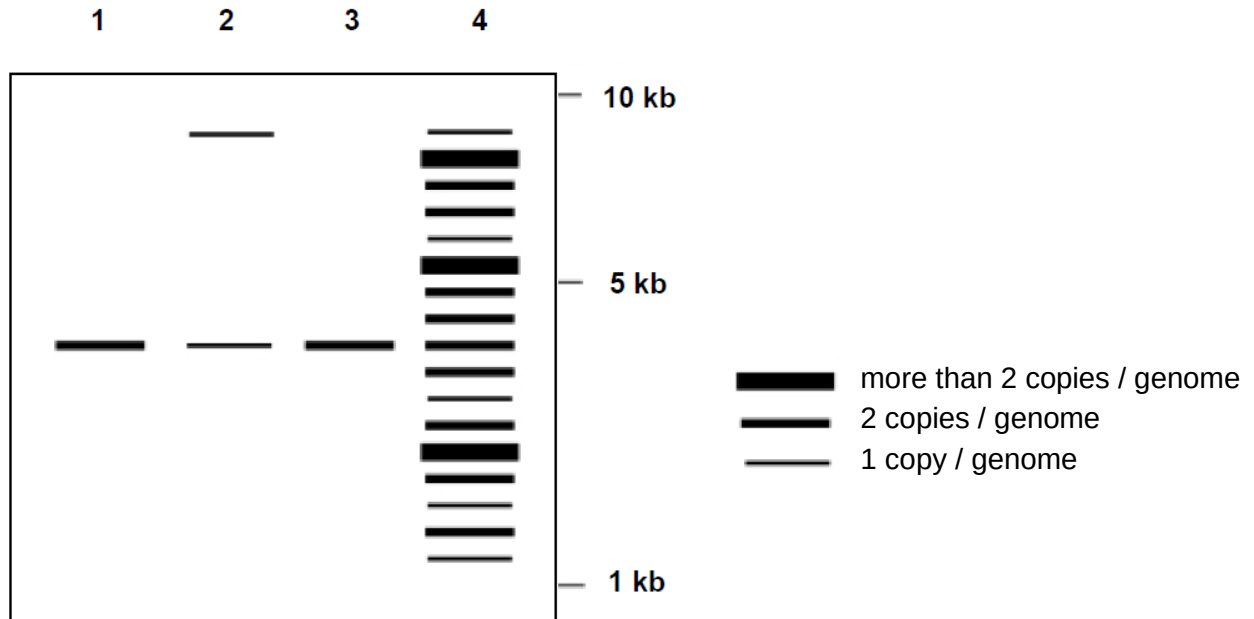
- A Individual bacteria that are not affected by antibiotics evolve resistance.
 - B Bacteria that are exposed to methicillin will become resistant to all antibiotics to which they are exposed.
 - C A number of individuals in a population of bacteria may be resistant to methicillin.
 - D A person who is resistant to methicillin will pass on the methicillin-resistant gene to the bacteria.
- 25 As part of the procedure to produce recombinant proteins in *E. coli*, you are asked to insert the gene encoding for the D-ONG protein into the pSEK-T vector. The restriction sites and selectable markers on the vector are shown below.



If the gene for D-ONG protein were to be inserted into the multiple cloning site, what should be added to the agar plate in order to screen for recombinant clones and how would the recombinant clones appear?

- | | Reagents to be added to agar plate | Appearance of bacterial colonies |
|---|------------------------------------|----------------------------------|
| A | X-gal only | White |
| B | X-gal and ampicillin | White |
| C | X-gal and tetracycline | Blue |
| D | Ampillicin only | Blue |

- 26 The *ras* oncogene is involved in a variety of human cancers. DNA isolated from a number of normal and cancerous human tissues were digested with *Eco*RI. A Southern blot was performed and the membrane was probed with radioactive-labelled cloned *ras* DNA. After exposure to x-ray film, the resulting autoradiograph is shown below.



Lane	Sample
1	white blood cells from a healthy human
2	lymphoma cells (cancerous)
3	bladder carcinoma cells (cancerous)
4	sarcoma cells (cancerous)

Based on the experimental results, which of the following statement(s) could be deduced?

- I. In tissue **3**, there was no mutation within the *ras* coding region.
 - II. The result for tissue **2** could be explained by separate point mutations that caused the removal of an *Eco*RI site and the creation of another within the probe DNA.
 - III. Gene amplification was involved in tissue **4**.
 - IV. In tissue **3**, the mutation did not affect existing *Eco*RI sites within the *ras* gene.
- A** I and II only
- B** II and III only
- C** III and IV only
- D** II and IV only

- 27** Which of the following are statements related to the goals for tackling the ethical, legal and social issues (ELSI) associated with the Human Genome Project?
- I. Examine the issues surrounding the completion of the human DNA sequence and the study of human genetic variation.
 - II. Examine issues raised by the integration of genetic technologies and information into health care and public health activities.
 - III. Examine issues raised by the integration of knowledge about genomics and gene environment interactions into non-clinical settings.
 - IV. Explore ways in which new genetic knowledge may interact with a variety of philosophical, theological, and ethical perspectives.
 - V. Explore how socioeconomic factors and concepts of race and ethnicity influence the use, understanding, and interpretation of genetic information, the utilization of genetic services, and the development of policy.
- A** I, II, III only
- B** I, II, IV only
- C** I, II, III, V only
- D** All of the above
- 28** Which feature of stem cells obtained from blood in the umbilical cord enables their use in the treatment of a variety of blood cancers?
- A** They are totipotent and can differentiate into any cells to replace the tissues affected by cancer.
- B** They are pluripotent and can replace blood stem cells affected by cancer.
- C** They are multipotent and can differentiate into any cellular component of blood.
- D** They are unipotent and can only differentiate into bone marrow cells.

- 29** Developing fish eggs can be treated to produce a diploid egg. In salmon, such eggs have been fused with haploid salmon sperm to give infertile triploid salmon.

Reproductive organ tissue from diploid trout was transplanted into newly hatched triploid salmon. This tissue matured as the fish grew and the salmon successfully produced viable trout sperm or eggs which resulted in young trout.

Which fish could be seen as both genetically identical and genetically modified?

- A** salmon providing eggs for treatment
 - B** trout providing reproductive organ tissue
 - C** young triploid salmon
 - D** young trout
- 30** Which of these processes could not increase crop yield?
- P** inserting genes for vitamin A production into rice
 - Q** inserting genes for pest resistance into cotton
 - R** inserting genes for herbicide resistance into oilseed rape
 - S** inserting genes for nitrogen fixation into soya beans
- A** P and S
 - B** Q and R
 - C** R and S
 - D** Q and P

2014 H1 Biology JC2 Preliminary Examination Paper 1 ANSWER KEY

1	A	2	D	3	B	4	B	5	D
6	C	7	B	8	A	9	D	10	C
11	A	12	D	13	C	14	B	15	A
16	A	17	C	18	A	19	B	20	B
21	D	22	A	23	D	24	C	25	B
26	C	27	D	28	C	29	D	30	A

A = 8

B = 7

C = 7

D = 8